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Section A

Answer **all** questions.

Refer to the separate Resource Folder to answer questions **1** and **2**.

1. (a) Dave says the distances of each leg in a long-distance triathlon are all double those in a middle-distance triathlon.

Tom says the distances of each leg in an Olympic triathlon are double those in a sprint triathlon.

Use data from **Table 1** to explain whether Dave or Tom is correct. [2]

....

....

....

....

(b) Piera says Jake breathed in less air per minute at 4 minutes than at 3 minutes because his breathing rate was lower.

Petula disagrees because Jake's airflow per breath was greater at 4 minutes than at 3 minutes.

Use data from **Table 2**, and the equation below it, to explain whether you agree with Piera or Petula. [2]

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(c) Use the information in **Table 4** and equations from pages 4 and 5 to answer the following questions.

(i) Calculate Malcolm's BMI. [2]

BMI = ...

(ii) Calculate Malcolm's maximum heart rate. [1]

maximum heart rate = ... bpm

(d) Karen wants to exercise in the aerobic zone. Use the information in **Tables 3** and **4** and on page 4 to determine the range in heart rate she should aim for. [1]

heart rate range is from .... bpm to .... bpm.

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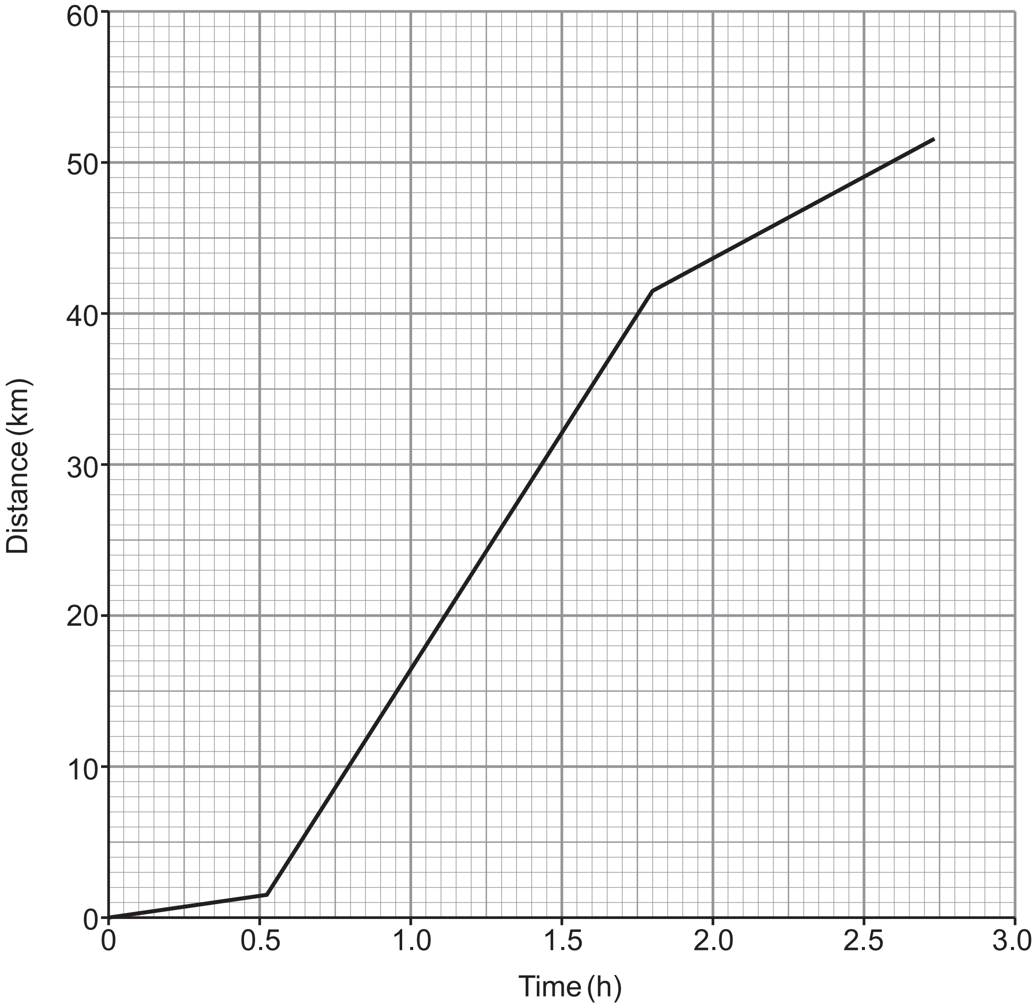
(e) Data in the **Triathlon times** section (page 5) and **Table 1** was used to plot distance-time graphs for each of the triathlons described. There were three sections in each graph to show the swim, cycle and run legs. The graphs did not include transition times. One of the graphs is shown below.



Determine which type of triathlon is represented by the graph and give **two** reasons for your answer. [3]

Type of triathlon: ....

Reason 1: ....

....

Reason 2: ....

....

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(f) Use the information about Triathlon times on page 5, the information in **Table 1** and an equation on page 7 to calculate the mean speed during an Ironman triathlon. [3]

mean speed = ... km/h

(g) When a triathlete mounts their bike they are travelling at 1.5 m/s.
Use the information in **Table 5** and an equation on page 7 to calculate the acceleration along level ground in a cycle leg. [3]

acceleration = ... m/s^2

(h) It is thought that the higher the age group of the triathlete, the lower the mean speed in the cycle leg. Use data from **Graph 1** to explain whether this statement is true for all age groups. [2]

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